

Kona Ya Hisabati

Complete UI/UX Design and Development Guideline

1. UI/UX Concept Overview

1.1 Purpose of the Platform

Kona Ya Hisabati is a web-based interactive mathematics learning platform designed to digitize the traditional Mathematics Learning Corner used in Tanzanian early grade classrooms. The platform aims to:

- Improve numeracy learning for Pre-Primary and Standard One to Two learners.
- Provide a child-friendly environment for interactive math practice.
- Equip teachers with structured digital activities and progress tracking tools.
- Support parents in guiding home-based practice numeracy learning.
- Ensure inclusive, accessible learning following UDL principles.

1.2 Target Users

1. **Children (Pre-Primary & Standard One-Two)**
 - Limited reading ability
 - Require visual, intuitive, highly interactive interfaces
 - Prefer audio guidance and large buttons
2. **Teachers**
 - Require structured lesson-aligned content
 - Need classroom resources, analytics, and printable materials
 - Require tools aligned with Tanzania's curriculum
3. **Parents / Caregivers**
 - Require simple navigation and explanations
 - Need practical home numeracy activities
 - Require video guides and low literacy support

1.3 Key Learning Principles

- **Early Childhood Education:** More visuals, less text; color-rich, playful interactions.
- **Universal Design for Learning (UDL):** Multiple means of engagement, representation, and expression.
- **Accessibility:** High contrast colors, simple navigation, audio instruction controls, alternative text, offline support.

2. Full UI Structure / Sitemap

2.1 Main Sections

1. **Home Page**
2. **Children Learning Activities**
 - Counting & Number Recognition
 - Shapes & Patterns
 - Addition & Subtraction
 - Math Games
 - Quizzes
 - Math Songs & Videos
3. **Teacher Support & Dashboard**
4. **Parent/Caregiver Support Section**
5. **About / Help / Settings**
6. **User Accounts / Login** (optional)

2.2 Sitemap (Hierarchical)

Home Page

- Start Learning
- Teacher Login
- Parent Guide
- About Kona Ya Hisabati
- Language Selector

Children Learning Section

- Counting & Number Recognition
 - Count Objects
 - Match Numbers
 - Number Ordering
- Shapes & Patterns
 - Identify Shapes
 - Create Patterns
 - Shape Sorting
- Addition & Subtraction
 - Single-digit addition
 - Single-digit subtraction
 - Story problems
- Math Games
 - Memory game
 - Matching game
 - Drag-and-drop games
 - Math Tag of war (mfano) https://www.instagram.com/reels/DUsl_8VATIr/
- Quizzes
 - Short assessments
 - Certificate printing
- Math Songs & Videos
 - Counting songs
 - Shape songs

- Animated story videos

Teacher Section/ Dashboard

- Lesson Plans
- Activity Library
- Learner Progress
- Printable Worksheets
- Classroom Tips

Parent Section

- How to Support Learning at Home
- Simple daily math tasks
- Video guides
- Progress view (optional)

2.3 User Flows

Child User Flow

Home → Start Learning → Choose Category → Interactive Activity → Reward/Progress → Continue or Exit

Teacher User Flow

Home → Teacher Login → Dashboard → Select Tools → Download/Track → Logout

Parent User Flow

Home → Parent Guide → View Tips → Watch Video → Access Home Activities

3. Detailed UI Descriptions for Each Page

3.1 Home Page

Purpose

Welcome page for all users; easy entry to learning activities.

Layout

- Top header: Logo, language toggle, teacher login
- Center: Main illustration (children learning)
- Large “Start Learning” button

- Secondary buttons: Teacher Area, Parent Guide
- Footer: About, Contact, Terms

Components

- Large rounded buttons
- High-resolution illustrations
- Top navigation bar

Sample Labels

- “Start Learning ”
- “Teacher Dashboard”
- “Parent Guide”

Interaction

- Hover: Light shadow expansion
- Click: Soft bounce animation
- Audio option: “Touch here to start learning!”

3.2 Learning Categories Page

Purpose

Provide children with icon based choices of math activities.

Layout

- Grid of colorful icons (6 main categories)
- Each icon shows an illustration (shapes, numbers, games)

Components

- Icon card (child friendly)
- Category title
- Simple audio prompts

Sample Labels

- “Counting,” “Shapes,” “Games,” “Quizzes”

Interactions

- Hover: Glow effect
- Click: Page transition with animation

3.3 Activity Page Template (All Learning Activities)

Purpose

Interactive page where children solve math tasks.

Layout

- Top bar: Home, Back, Audio
- Middle: Interactive component (drag-and-drop, buttons, animations)
- Bottom: Next Activity button

Components

- Large interactive element
- Object illustrations (fruit, toys, animals)
- Reward animations (stars)

Sample Text

- “Count the apples”
- “Touch the correct number”

Interaction Behavior

- Click feedback: Color change
- Drag feedback: Snap-to-place
- Correct answer: Star animation and voice “Good job!”

4. Learning Activity UI Templates

4.1 Counting & Number Recognition

- Drag objects to match numbers
- Select correct number from 1–10
- Moving number line

4.2 Shapes & Patterns

- Identify shapes by tapping
- Complete patterns using drag-and-drop
- Shape matching memory game

4.3 Addition & Subtraction

- Visual math (apples added/removed)

- Simple equations ($3 + 2 = ?$)

4.4 Math Games

- Maze game (collect numbers)
- Shape-hunt adventure

4.5 Quizzes

- Timed tasks
- Star-rating scorecards

4.6 Math Songs & Videos

- Video player with large controls
- Playlist of math songs

5. Teacher Dashboard UI Requirements

5.1 Content Access

- Clear folder-like structure
- Curriculum-aligned activities

5.2 Progress Tracking

- Charts (bar/line)
- Class-level summaries
- Downloadable reports

5.3 Materials

- PDF worksheets
- Flashcards
- Lesson guides

5.4 Classroom Tools

- Tips for differentiation
- UDL strategies
- Remedial plans

6. Parent/Caregiver Support UI

Key Elements

- Step-by-step guides
- Simple video tutorials
- Printable home activities
- Low-literacy-friendly icons
- Large fonts

7. Style Guide / Design System

7.1 Color Palette

- Primary Blue:
- Primary Yellow:
- Secondary Green:
- Orange (high interest)
- Background Light:

7.2 Typography

- Headings: Poppins
- Body text: Nunito

7.3 Buttons

- Rounded corners
- Height: 70–90px for children
- Bold labels

7.4 Icons

- Flat, colorful, child friendly
- Use culturally appropriate visuals

7.5 Spacing

- 16px baseline grid
- Clear padding around elements

8. Accessibility & Inclusivity Requirements

- High contrast colors
- Alt-text for images
- Audio instruction button on all activities
- Touch-friendly 1.5cm minimum tap-target
- Simple language (Kiswahili & English)
- Dyslexia-friendly mode (optional)

9. Flow Diagrams (Text-Based)

10.1 Child Flow

Home → Start → Select Category → Activity → Reward → Next Activity → Exit

10.2 Teacher Flow

Home → Login → Dashboard → Select Tool → Track/Download → Logout

10.3 Parent Flow

Home → Parent Guide → Watch Video → Try Home Activity → Repeat